

Electrochemotherapy

Electrochemotherapy is used to treat cancers that have spread to the skin.

On this page

- [What is electrochemotherapy?](#)
- [When is electrochemotherapy used?](#)
- [Having electrochemotherapy](#)
- [Side effects of electrochemotherapy](#)
- [About our information](#)
- [How we can help](#)

What is electrochemotherapy?

[Chemotherapy](#) uses anti-cancer (cytotoxic) drugs to destroy cancer cells. Electrochemotherapy uses chemotherapy and a small electrical current to treat cancer cells.

The doctor injects a low dose of chemotherapy into the tumour or a vein (intravenously).

After this they put a probe (electrode) directly over the tumour to give an electrical pulse. The pulse changes the outside layer of the cells. This is called electroporation. It helps the chemotherapy get into the cancer cells and destroy them.

When is electrochemotherapy used?

Electrochemotherapy may be used to treat skin tumours caused by [secondary breast cancer](#).

It is sometimes used to treat cancer that has spread to an area of skin, including [advanced melanoma](#).

It is also sometimes used to treat [skin cancer](#) that has spread.

This treatment can help to control symptoms such as:

- bleeding
- ulcers
- pain.

Some people will have the treatment more than once.

Electrochemotherapy is a specialised treatment and is not available at all hospitals. Your cancer doctor or nurse can explain more about your area.

Having electrochemotherapy

You usually have electrochemotherapy as an outpatient and go home the same day. Some people may stay in hospital overnight.

If you are having treatment to a single area, the doctor will give you a local anaesthetic to numb the area. They use a fine needle to inject the anaesthetic into the skin.

If you are having treatment for lots of areas at the same time you may have a general anaesthetic.

You will have chemotherapy in one of these ways:

- As an injection into the skin being treated after the anaesthetic.
- As an injection into a vein. The doctor puts a short, thin tube called a [cannula](#) into a vein in your arm or the back of your hand. They give the chemotherapy drug through the cannula.

After the anaesthetic, the doctor puts a probe (electrode) directly over the tumour to give an electrical pulse. The electrical pulse may make nearby muscles tighten suddenly or twitch without your control. This side effect can be uncomfortable. Your doctor may be able to change the position of the probe to help. This side effect will stop after the treatment finishes.

You may feel some discomfort in the skin after treatment. Your doctor may advise you to take mild painkillers to help. The treated area is usually covered with a dressing to protect it while it heals.

Side effects of electrochemotherapy

The side effects of electrochemotherapy are different to side effects of chemotherapy. This is because a very low dose is used. Most side effects affect the area that is being treated.

Your nurse or doctor will give you more information about side effects that may affect you.

You may get some mild sickness or feel breathless.

You may also get side effects in the treated area. These can include:

- redness and swelling
- mild scarring
- the skin in the area may become lighter or darker - this might be permanent.

You may have mild pain for a few weeks. Your doctor can prescribe painkillers to take until it goes away.

Your nurse will tell you how to look after the area before you go home. The treated area often develops a scab which usually heals over a few weeks.

Contact your doctor or nurse straight away if you have signs of infections such as:

- the area becomes red, hot or swollen
- if there is any discharge.

You may need treatment with antibiotics.

About our information

References



Below is a sample of the sources used in our melanoma information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

Michielin O, van Akkooi ACJ, Ascierto PA, et al. Cutaneous melanoma: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. *Annals of Oncology*. 2019; 30, 12, 1884-1901 [accessed May 2022].

Michielin O, van Akkooi ACJ, Ascierto PA, et al. ESMO consensus conference recommendations on the management of locoregional melanoma: under the auspices of the ESMO Guidelines Committee. *Annals of Oncology*. 2020; 31, 11, 1449-1461 [accessed May 2022].

Peach H, Board R, Cook M, et al. Current role of sentinel lymph node biopsy in the management of cutaneous melanoma: A UK consensus statement. *Journal of Plastic, Reconstructive & Aesthetic Surgery*. 2020; 73, 1, 36-42 [accessed May 2022].

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, Professor Samra Turajlic, Consultant Medical Oncologist.

With thanks to: Dr Stephanie Arnold, Consultant; Kerry Jane Bate, Advanced Nurse Practitioner; Dr Philippa Closier, Clinical Oncologist; Sharon Cowell-Smith, Macmillan Advanced Nurse Practitioner Skin Cancers; and Dr Benjamin Shum, Medical Oncologist.

Thanks also to the other professionals and people affected by cancer who reviewed this edition, and to those who shared their stories.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

Reviewed: 01 July 2023 | Next review: 01 July 2026

⚠ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help